

XSA continuity (strict) runs

Introduction

Extended survival analysis (XSA) (Shepherd, 1992) is a tuned virtual population analysis (VPA) that uses information on individual cohort sizes to estimate survivors at each age in the terminal population. XSA estimates the survivors (i.e. terminal Ns by age and year) for each observed value of CPUE. This is done by calibration regression to predict population numbers-at-age by year for each CPUE series and then projecting along the cohort to the oldest age or most recent year. In addition shrinkage to the mean is performed, where the terminal Ns also including a term related to recent Fs or Fs at younger ages (shrinkage to the mean F) and numbers-at-age for recruiting age classes are estimated from the geometric mean of recent recruitments (shrinkage to the mean n). Time series weights can be applied to discount past values. The estimated survivors are then used to initiate a modified iterated VPA. XSA has been used for formulating advice in the previous assessments of the Mediterranean swordfish stock.

Materials and Methods

The age structured assessment was conducted using XSA in R using the FLXSA package (part of the FLR-project, Kell et al., 2007; <http://www.flr-project.org/>). The available CAS data covered the period 1985-2018 and the catch at age (CAA) data were generated using a statistical mixture distribution analysis similarly to the previous (2016) assessment. The available CAS did not justify splitting CAS into ages greater than 6 and in line with the previous assessment XSA runs were conducted with a plus group of 6. As in 2016 continuity assessment, natural mortality was assumed to equal 0.2 for all ages and years. In the 2016 assessment it was assumed that fish first mature at age 3 (when 50% are mature) and are fully mature at older ages. Considering new information (ICCAT 2020), it was currently assumed 15% and 65% maturation at ages 2 and 3 respectively. Full maturity was considered from age 4 onwards. Weights-at-age were derived from the mixture analysis and were consistent with the CAA. Figures 1 and 2 illustrate the available CAA data by age and year.

Three standardized CPUE series were considered as in the 2016 assessment: Greek longliners (SCRS 2020/021), Moroccan longliners (SCRS 2020/026) and Spanish longliners (SCRS 2020/043). The standardized CPUE indices were not differentiated by age and were considered to be representative of the 2-4 age-group abundances. Fleet catchability was assumed to be independent of year-class size for all terminal years.

XSA control settings

Settings were similar to those adopted in 2016 and are as follows:

shk.n=TRUE, shk.yrs=10, rage=1, maxit=100, shk.f=TRUE, shk.ages=2, fse=0.5, min.nse=0.3, qage=6, window=100, tsrange=10, tspower=1, vpa=TRUE

F range: minfbar=2, maxfbar=4

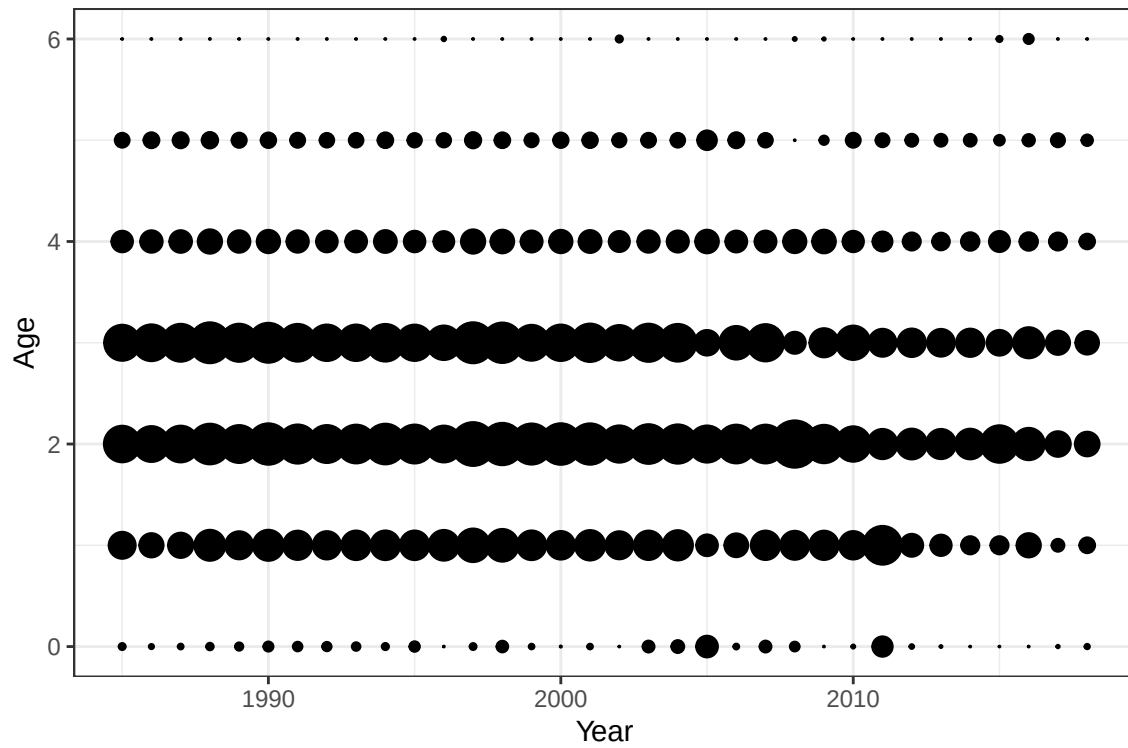


Figure 1. Catch-at-age by year. Bubbles are proportional to abundance.

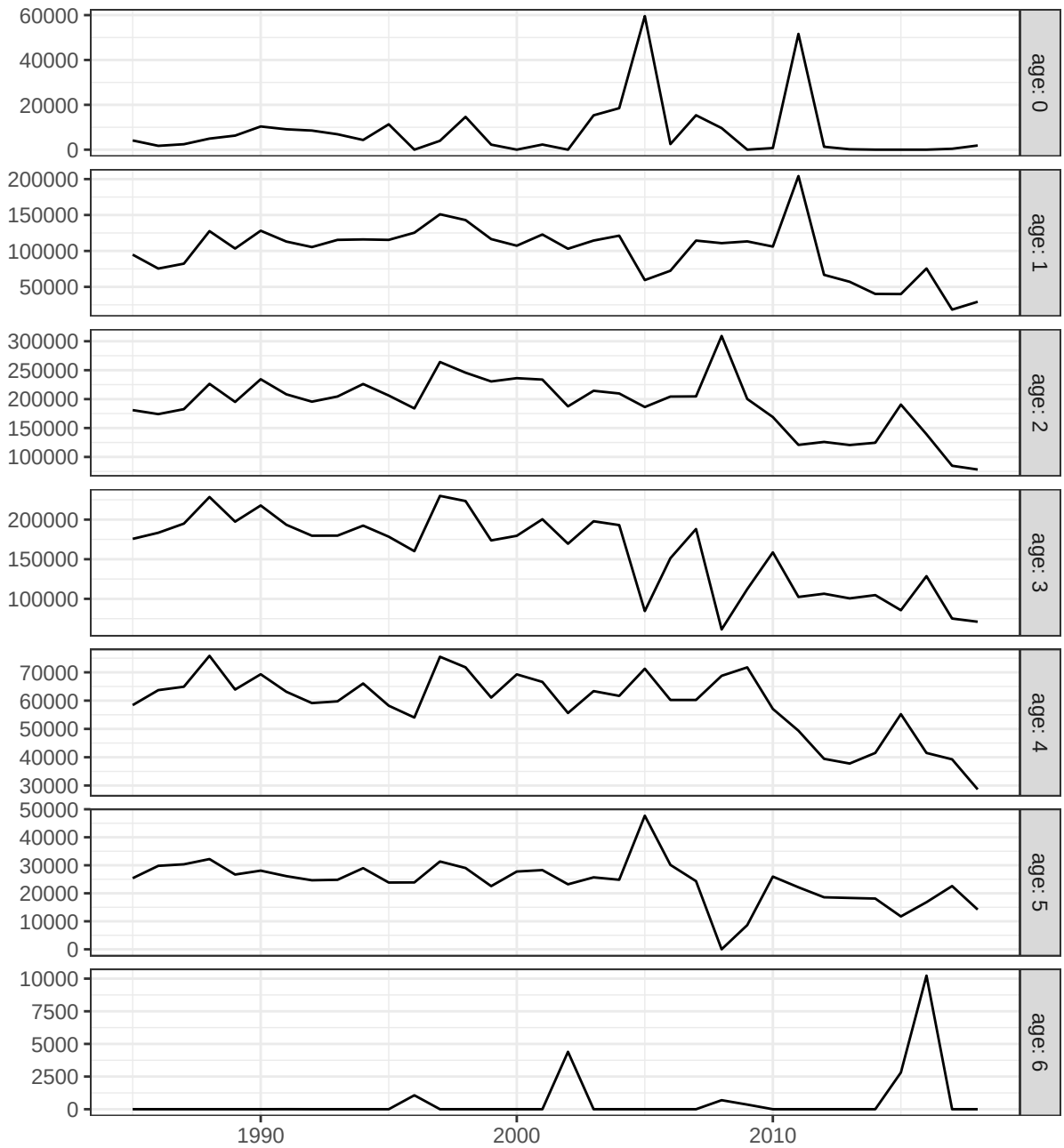


Figure 2. Time series of catch-at-age.

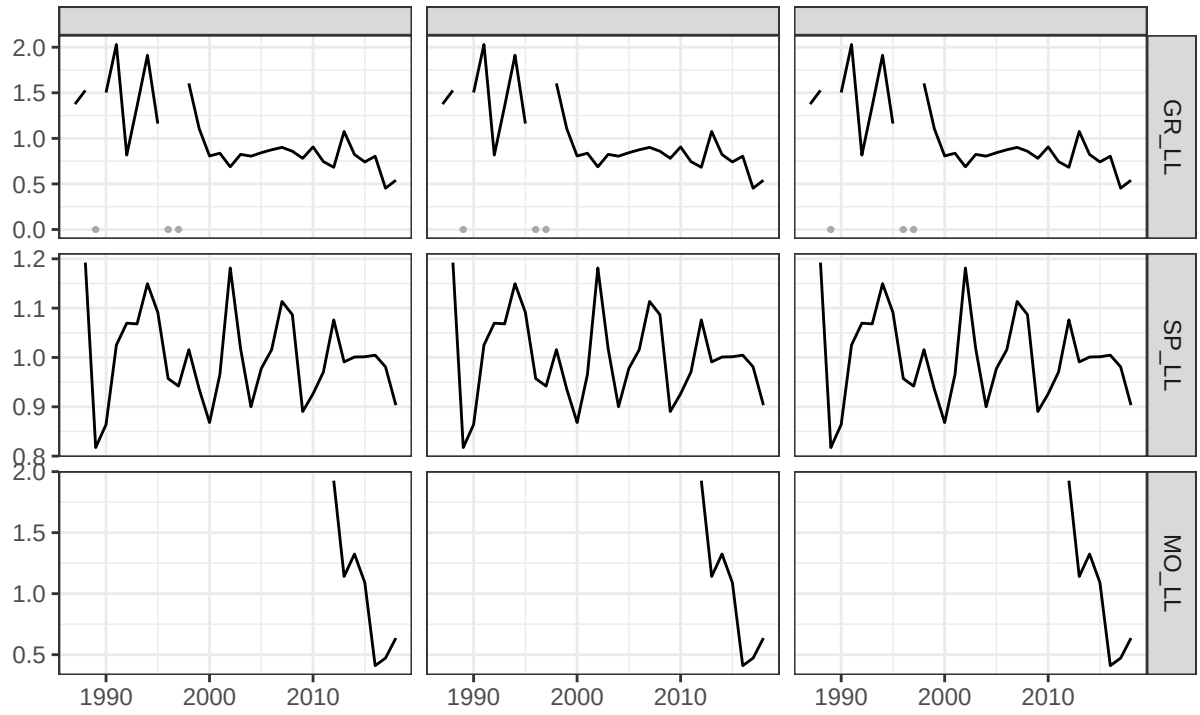


Figure 3. Time series of CPUEs.

Results

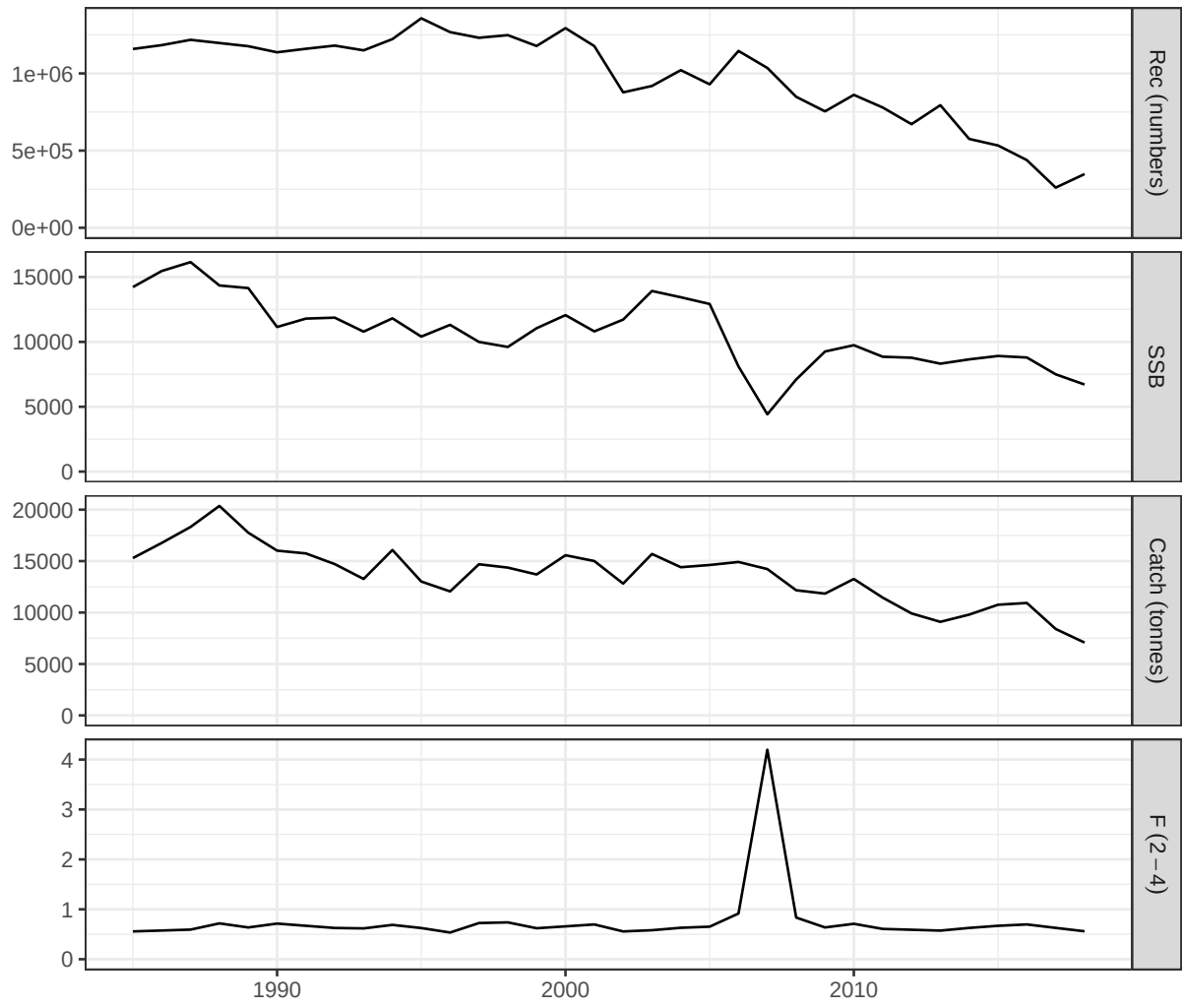


Figure 4. XSA time series estimates

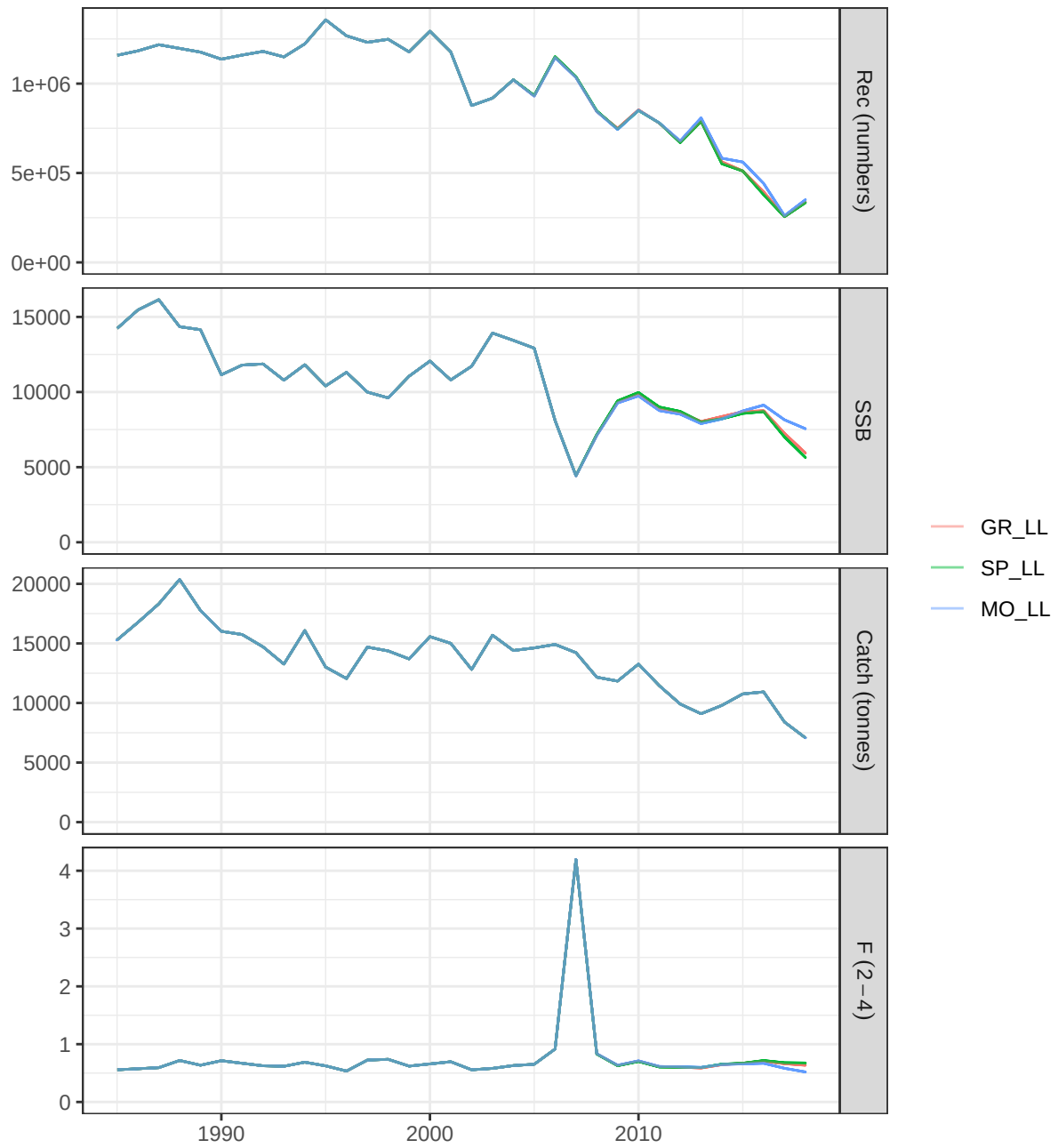


Figure 5. XSA time series estimates by CPUE

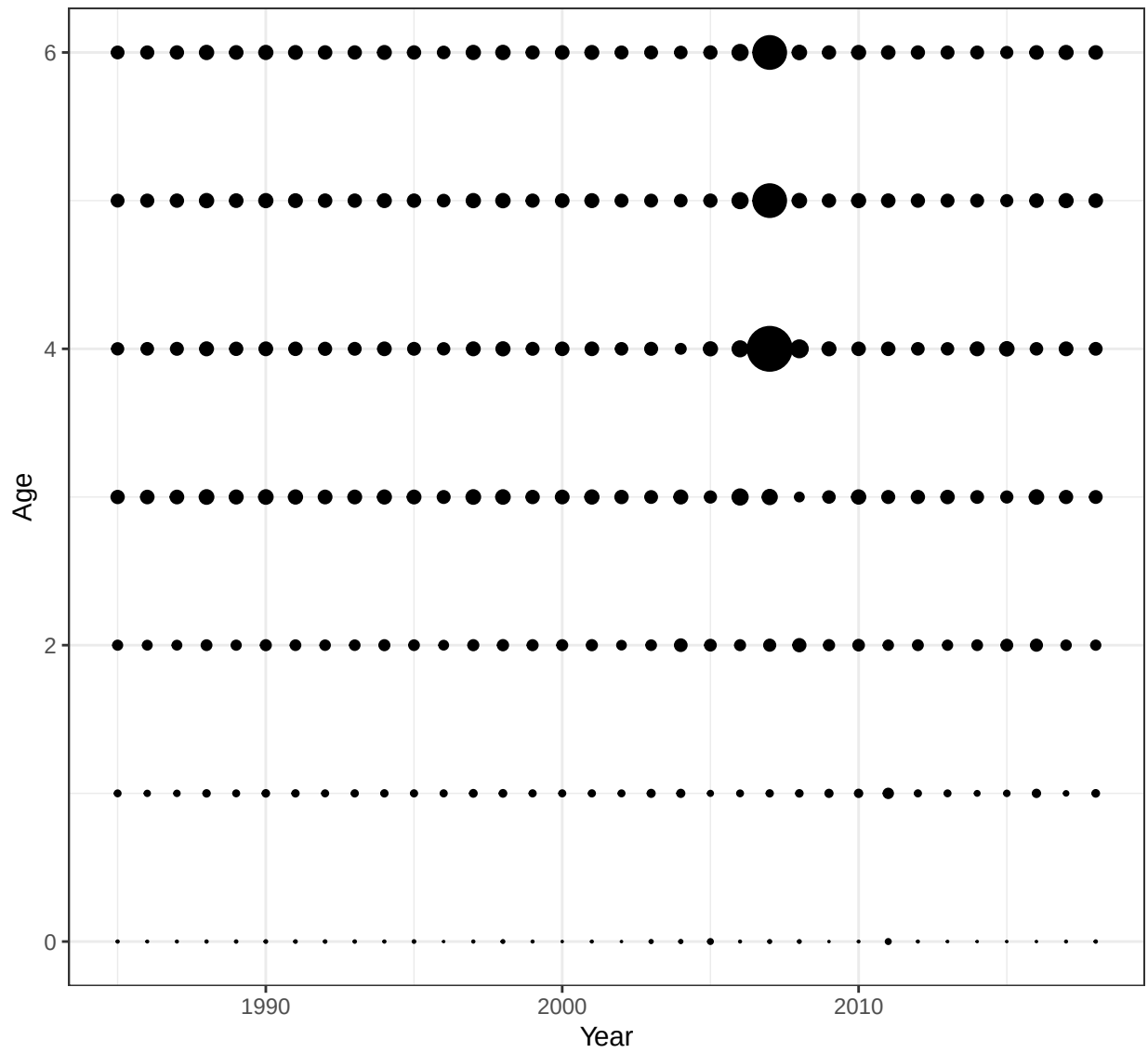


Figure 6. Estimates of harvest-at-age by year. Bubbles are proportional to the estimated rates.

Diagnostic plots

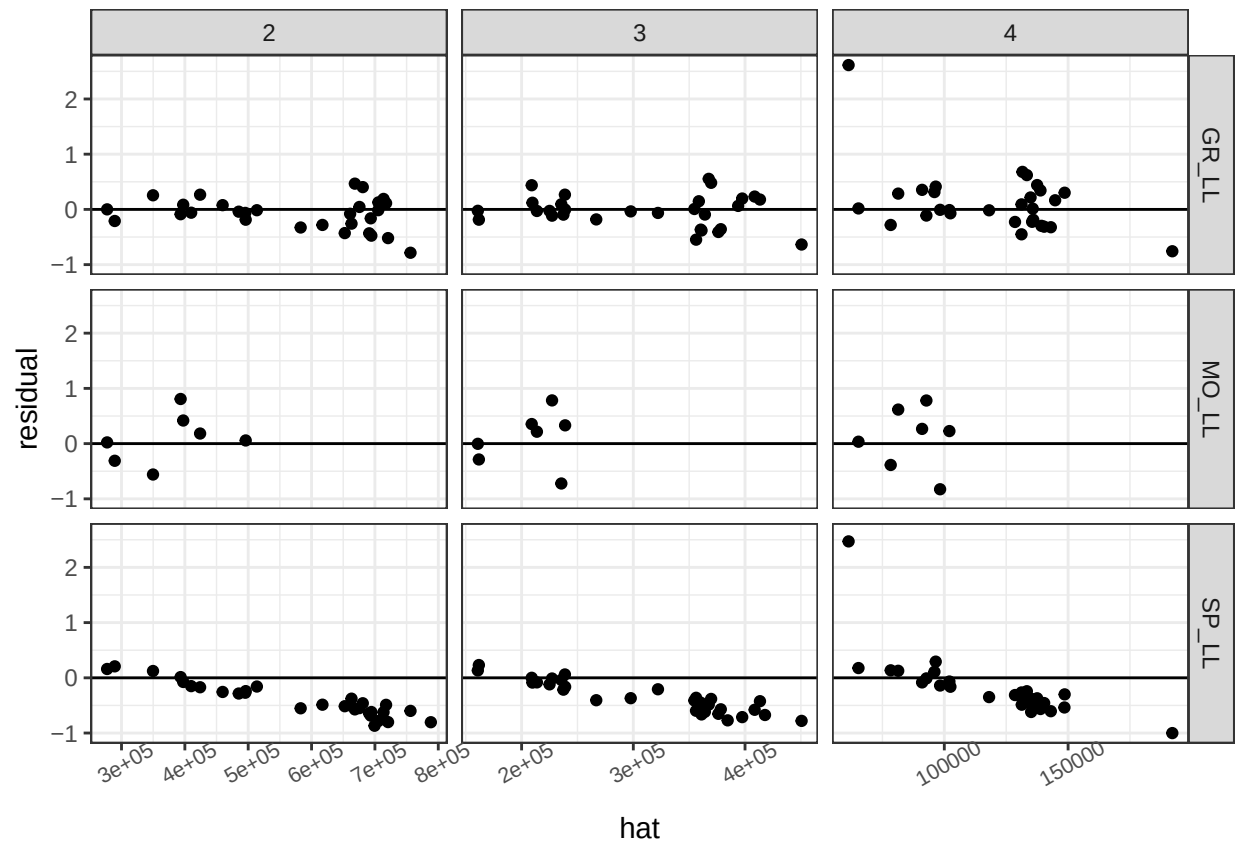


Figure 7. Residuals against fitted value.

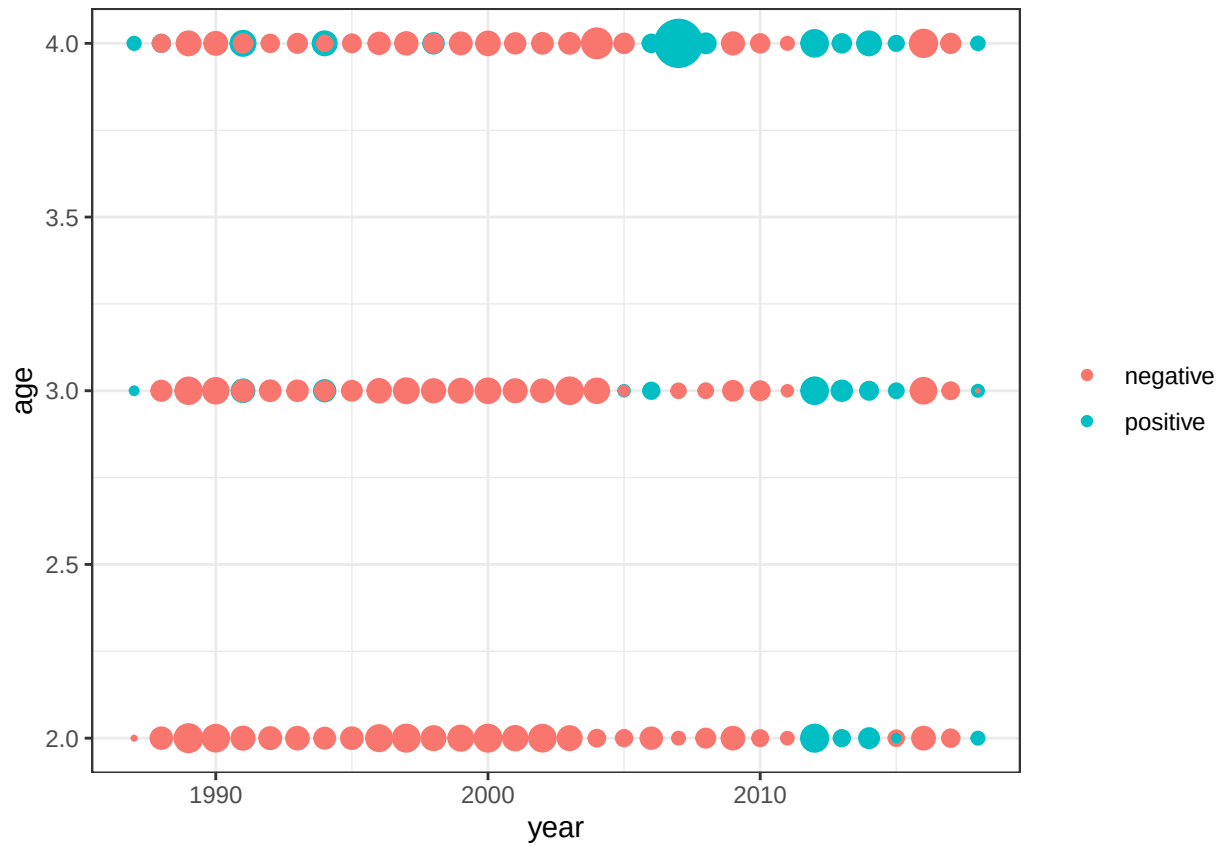


Figure 8. Residuals against year.

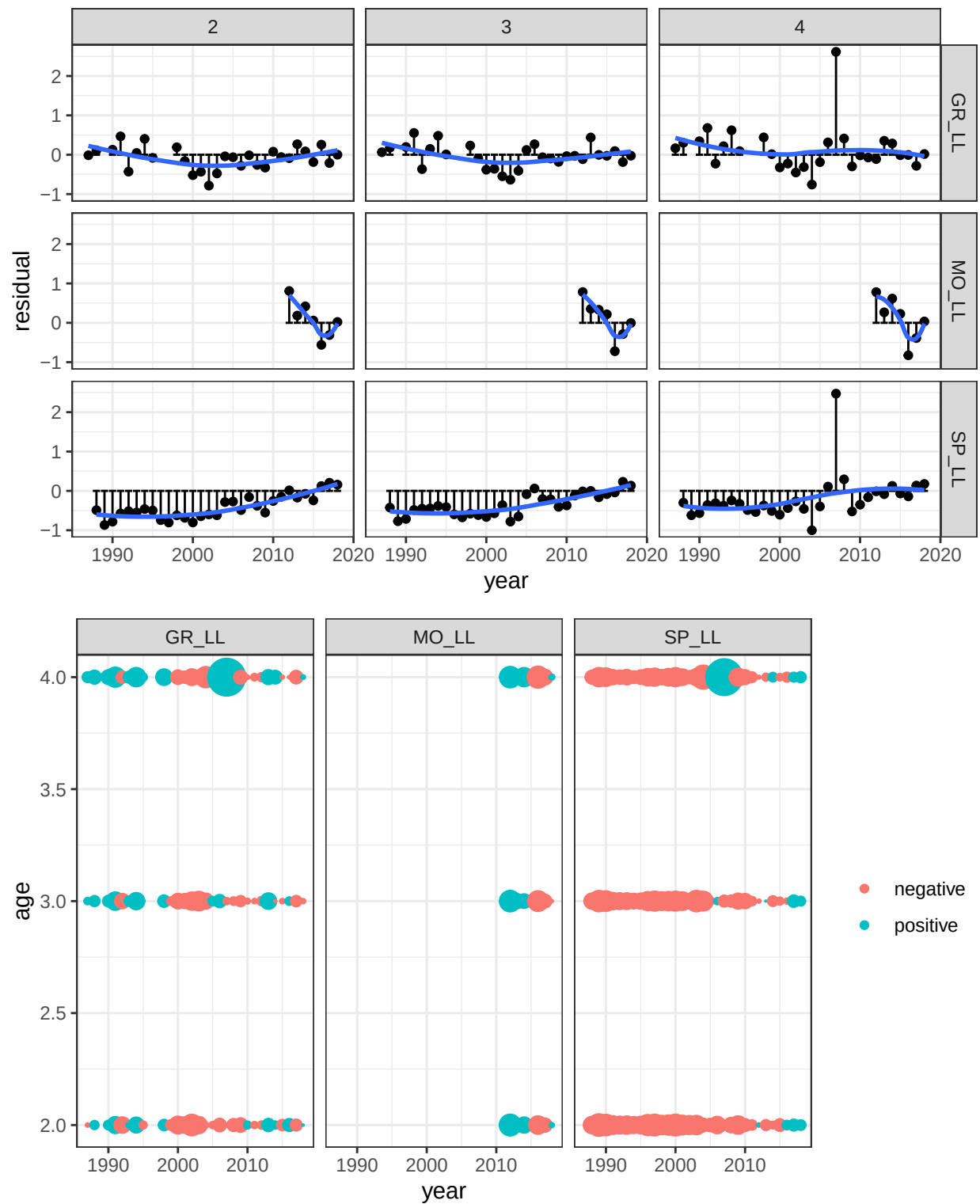


Figure 9. Residuals by fleet against year.

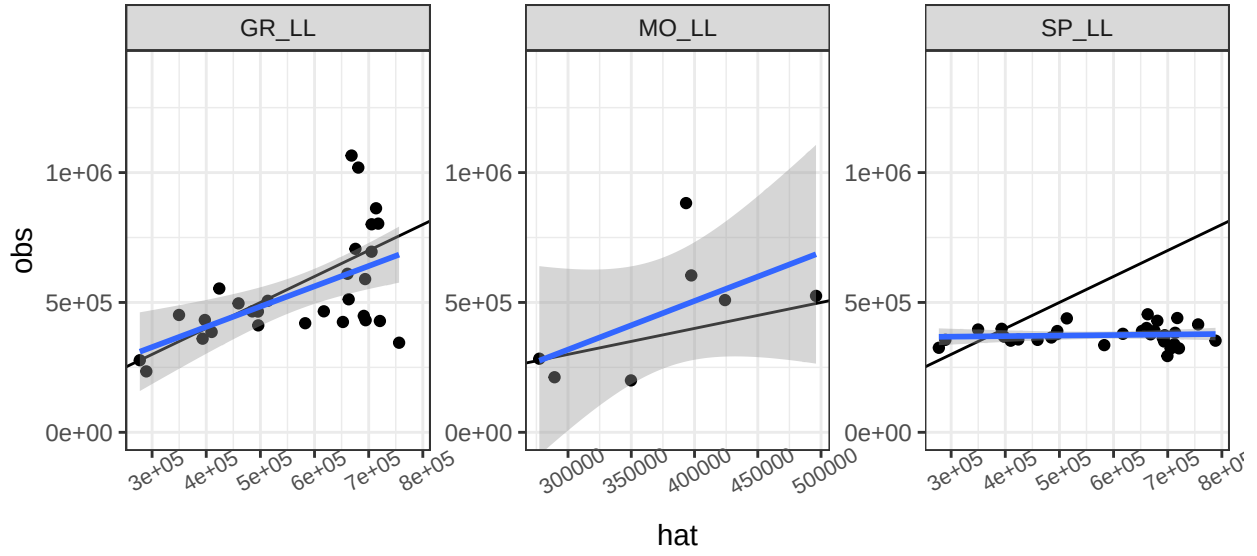


Figure 10. Calibration regression plots for age 2.

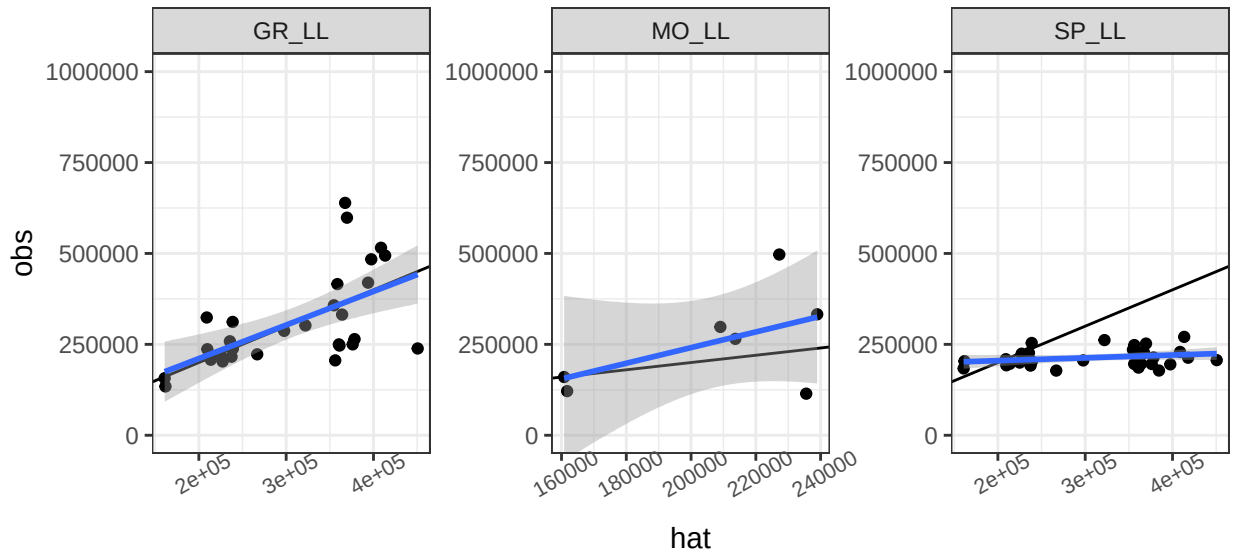


Figure 11. Calibration regression plots for age 3.

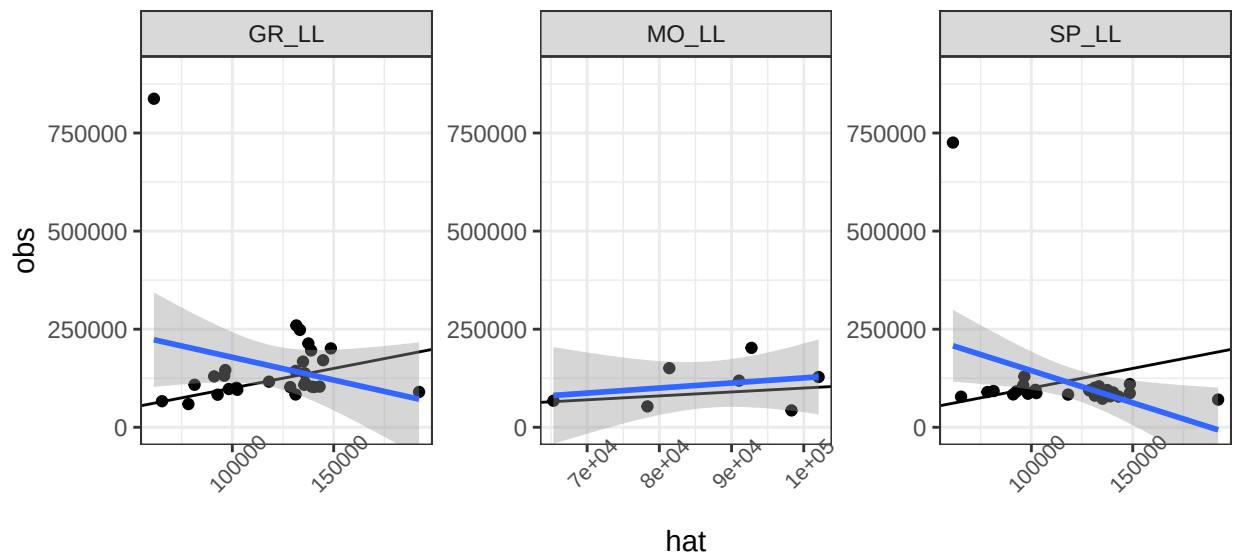


Figure 12. Calibration regression plots for age 4.

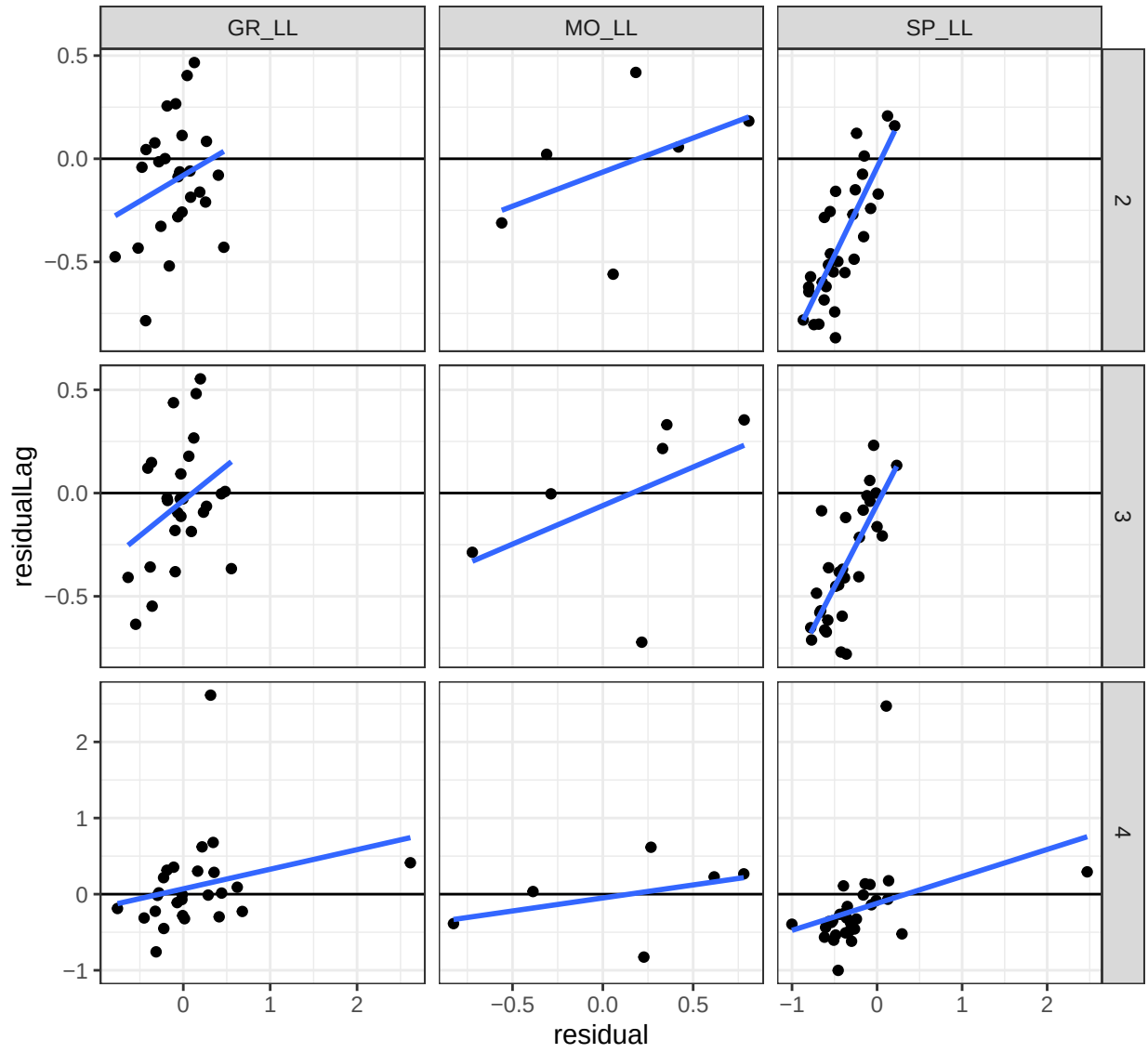


Figure 13. AR plots of lagged residuals

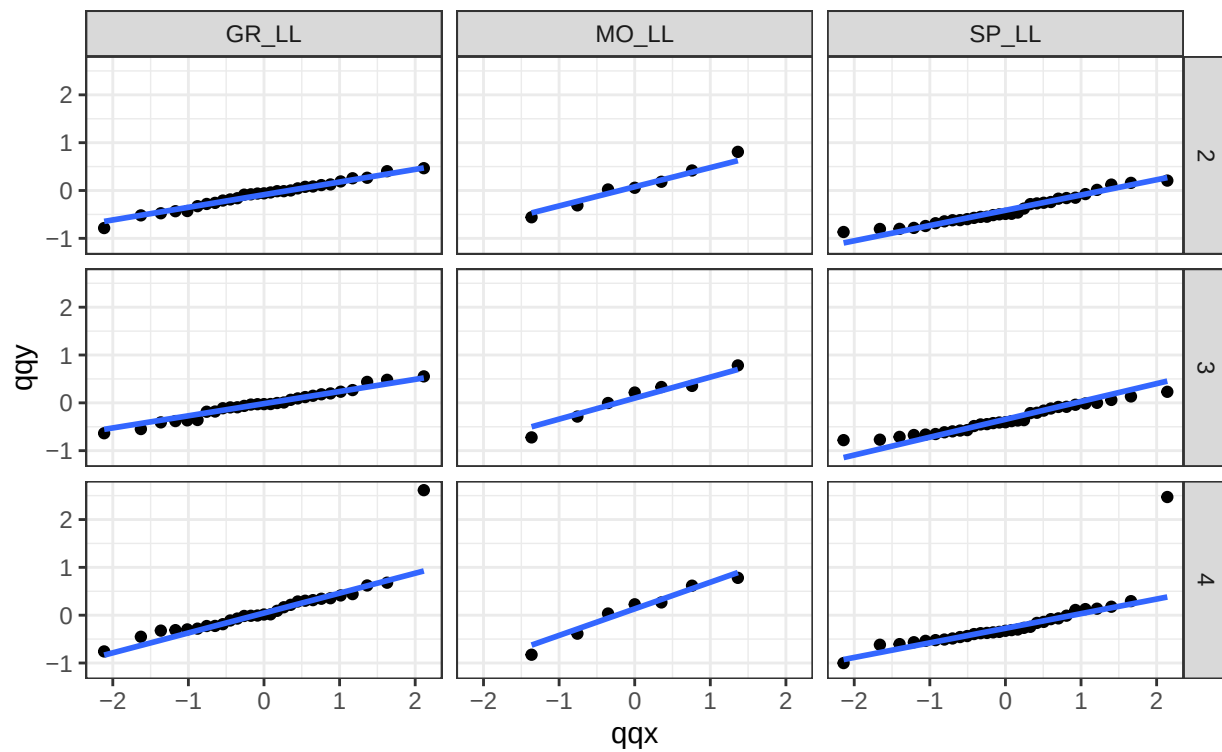
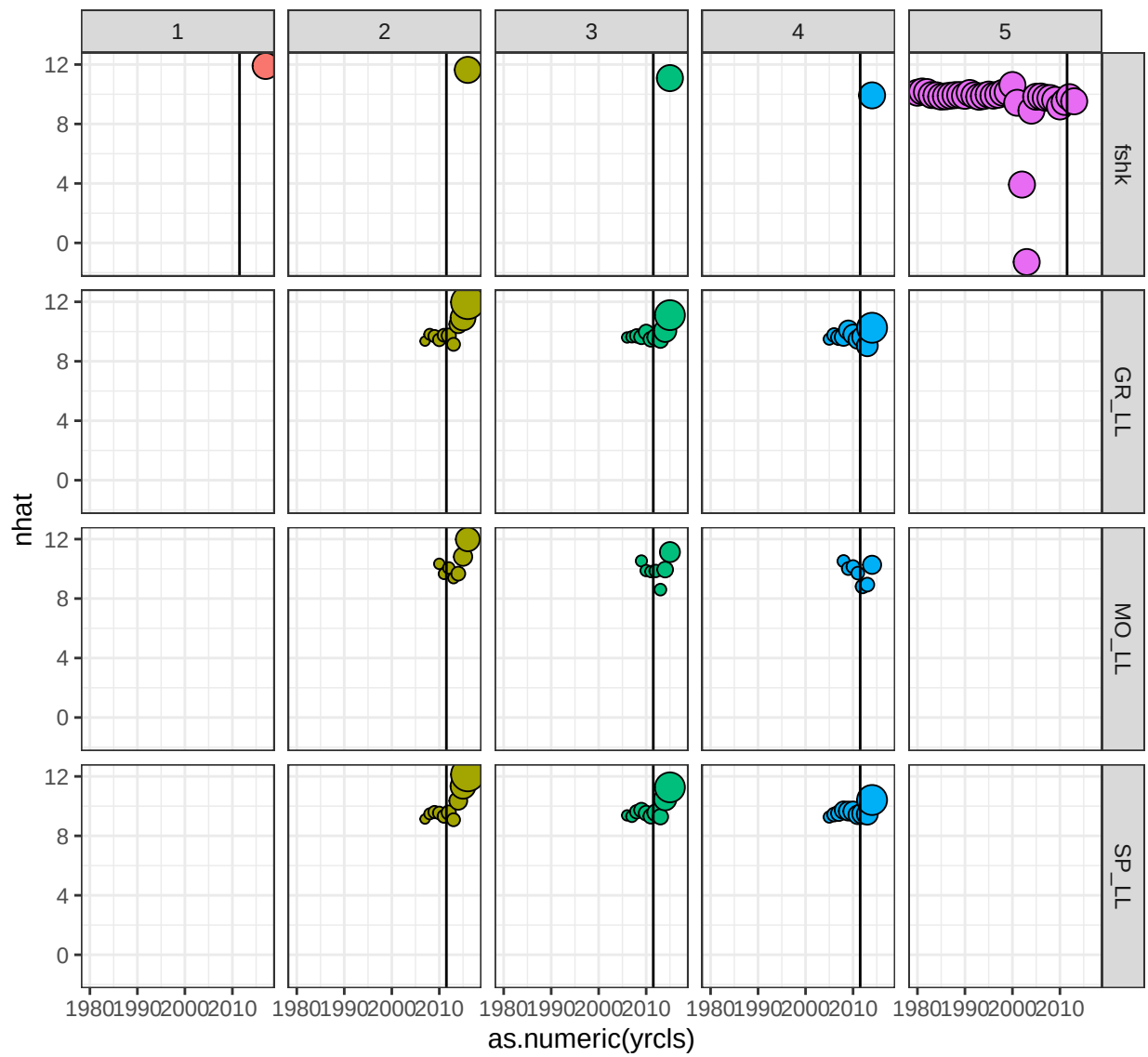
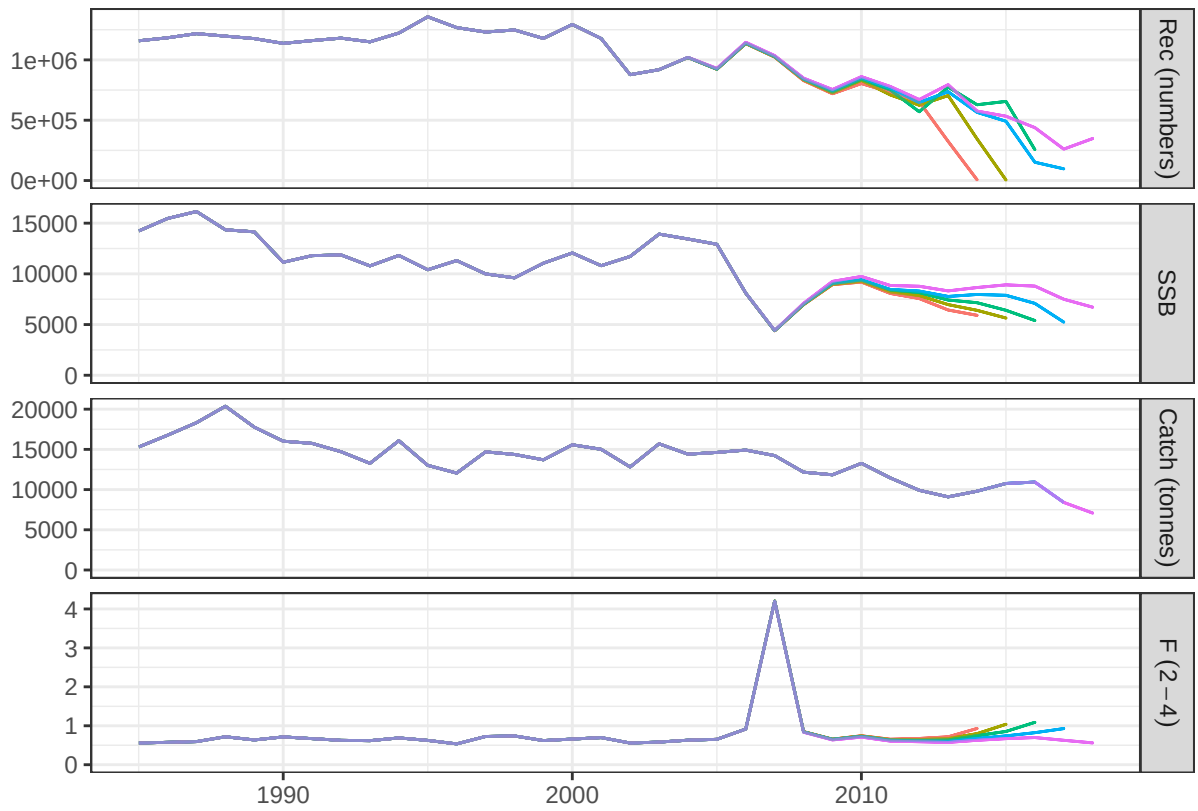


Figure 14. QQ plots to check for normality



***Figure 15.** Weights for terminal year Ns for each CPUE observation and shrinkage



***Figure 16.** Retrospective plot

References

ICCAT, 2020

Kell, L.T., Mosqueira, I., Grosjean, P., Fromentin, J.M., Garcia, D., Hillary, R., Jardim, E., Mardle, S., Pastoors, M.A., Poos, J.J. and Scott, F., 2007. FLR: an open-source framework for the evaluation and development of management strategies. *ICES Journal of Marine Science: Journal du Conseil*, 64(4), pp.640-646.

SCRS 2020/021

SCRS 2020/026

SCRS 2020/043

Shepherd, J. (1999). Extended survivor's analysis: An improved method for the analysis of catch-at-age data and abundance indices. *ICES Journal of Marine Science*, 56(5), 584–591.